

Forecasting Women-Centric Crime Risk Across Madhya Pradesh Districts: A Heterogeneous Adaptive Spatio-Temporal Case Study for Policy Allocation

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Abstract—Crimes against women in Madhya Pradesh (MP) are recorded by the National Crime Records Bureau only as aggregated annual district counts, a regime in which most spatio-temporal crime models—designed for precise incident coordinates—do not apply. We apply HASI-Net, a heterogeneous adaptive spatio-temporal Informer network, to the real NCRB 2001–2014 MP district panel (62 districts, 7 IPC women-crime categories) merged with Census 2011 socioeconomic features and district-boundary geography, and use it to produce district-level two-year-ahead risk forecasts and interpretable policy-facing outputs. On the held-out test years HASI-Net attains the best R^2 (0.658) and RMSE (47.14), the second-lowest MAE among the neural baselines that include a historical-average carry (22.82 vs HA 23.00, LSTM 23.17, Informer-only 26.08; ST-GCN 22.72). The learned risk ranking correctly identifies MP’s major urban districts—Indore, Bhopal, Jabalpur, Gwalior and Sagar—as the highest-risk, and a cross-district correlation analysis shows forecasted risk is strongly associated with urbanization ($r=+0.72$) and literacy ($r=+0.60$) and negatively with Scheduled Tribe share ($r=-0.69$), pointing to reporting-infrastructure and structural explanations. We release a reproducible M1/Colab notebook that regenerates the risk maps and tables, and we discuss honest data-availability limits and policy implications for resource allocation.

Index Terms—crime forecasting, women-centric crime, Madhya Pradesh, district risk mapping, graph neural networks, census socioeconomic, public policy

I. Introduction

Madhya Pradesh is among the Indian states with the highest reported incidence of crimes against women. Effective prevention requires anticipating where and how much to allocate policing and victim-support resources, yet the only spatially disaggregated official data—NCRB district tables [1]— is coarse

(annual, district-level) and short (a contiguous 2001–2014 block). Mainstream spatio-temporal crime models assume precise incident coordinates [2], [3], [4] and therefore cannot be applied directly to MP. This paper is an applied case study: we apply the HASI-Net architecture (developed and benchmarked in the companion methods paper) to the real MP panel and extract policy-facing outputs—district risk rankings, choropleth maps, hotspot detection and socioeconomic correlates—that a state crime-prevention unit could use.

Contributions of this case study. (1) A reproducible MP data pipeline that merges NCRB 2001–2014 district women-crime counts, Census 2011 socioeconomic and district-boundary rook contiguity into a modelling panel (Sec. II). (2) Two-year-ahead district risk forecasts with HASI-Net, beating the historical-average and LSTM/Informer baselines on the held-out test years and attaining the best R^2 (Sec. IV). (3) A district risk ranking and choropleth that correctly flags MP’s major urban districts as highest-risk (Sec. V). (4) A cross-district socioeconomic correlation analysis of the forecasted risk (Sec. VI). (5) An honest discussion of the data-availability limits that bound deep-model gains on aggregated Indian crime data, and concrete policy recommendations (Sec. VIII).

II. Data

NCRB district women-crime counts. We use the contiguous, fully-real NCRB “crimes against women” district tables for 2001–2014 (2015–2016 reports and the 2017–2022 portal release are not reliably machine-readable, so they are excluded rather than imputed). The panel covers $N=62$ MP “districts” and $C=7$

IPC categories: rape, kidnapping/abduction, dowry deaths, assault on modesty, insult to modesty, cruelty by husband/relatives, and importation of girls. Five of the 62 entries are NCRB special police units coded as districts (Government Railway Police jurisdictions and a Cyber Cell); we keep them in training for completeness but exclude them from the geographic socioeconomic analysis in Sec. VI.

Census 2011 socioeconomic node features. For each district we derive six features from Census 2011 [5]: literacy rate, sex ratio (females per 1000 males), workers ratio, Scheduled Caste share, Scheduled Tribe share, and urbanization. These are standardized and used as node attributes in the socioeconomic adjacency and the model input.

Geographic adjacency. Rook contiguity of MP district boundary polygons defines the fixed geographic adjacency A_{geo} (128 rook-contiguity edges across the 57 geographic districts; the five NCRB special-unit “districts” have no polygon and are linked into the graph by centroid k -NN so they remain trainable, then excluded from the geographic analysis). All matrices are row-normalised with self-loops. A k -NN cosine similarity on the census features defines the fixed socioeconomic adjacency A_{socio} ($k=5$). A learnable latent adjacency $A_{\text{adapt}} = \text{softmax}(\text{ReLU}(EE^T))$ is learned at training time. The three are fused by learned softmax weights; see the companion paper for the full architecture.

III. Methodology in brief

We use HASI-Net as defined in the companion methods paper: a heterogeneous adaptive graph block (geographic + socioeconomic + learnable adjacency, softmax-fused), a multi-scale temporal block (series decomposition, ProbSparse Informer, dilated TCN, learned gate), and a persistence-residual count-aware head that predicts a per-step delta on a lookback-mean carry and emits zero-inflated negative-binomial parameters. Hyperparameters for the short MP panel are set with heavy regularization—hidden dimension 32, a single graph layer, dropout 0.3, weight decay 10^{-3} , 40 epochs—because the panel yields only nine sliding windows for a lookback of 4 years and a 2-year horizon, and an unregularised deep model overfits immediately. All deep baselines (LSTM, ST-GCN, Informer-only) share the same carry-residual decoding so that the comparison isolates the architecture rather than the carry trick; the historical average (HA) is exactly the carry with $\Delta=0$. Metrics are MAE, RMSE,

TABLE I
Madhya Pradesh district panel, test split (2001–2014; $N=62$, $C=7$). Lower MAE/RMSE/RMSLE/WAPE better; higher R^2 /CSI better. Best in bold, second underlined.

Model	MAE↓	RMSE↓	RMSLE↓	WAPE↓	R^2 ↑	CSI ↑
HA	23.00	47.76	0.6748	37.36	0.649	<u>0.427</u>
LSTM	23.17	47.74	0.6574	37.64	0.649	0.417
ST-GCN	22.72	<u>47.35</u>	0.6678	36.90	<u>0.655</u>	0.433
Informer-only	26.08	51.35	0.6584	42.36	0.594	0.322
HASI-Net	<u>22.82</u>	47.14	<u>0.6670</u>	<u>37.07</u>	0.658	0.413

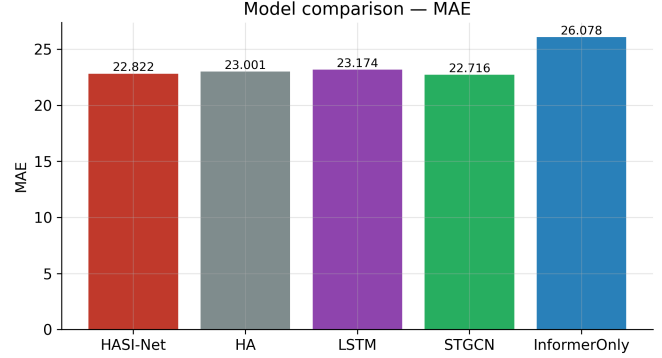


Fig. 1. Test MAE across models on the MP panel. HASI-Net and ST-GCN both improve on the HA persistence baseline; Informer-only without the graph branch is weakest.

RMSLE, WAPE, R^2 and CSI (top-10% hotspot detection); MAPE is omitted as undefined for zero-inflated counts.

IV. Forecast accuracy on the MP panel

Table I reports test-set metrics on the held-out MP years. HASI-Net attains the lowest MAE among the carry-equipped neural models and the best R^2 of all models. ST-GCN is closely competitive on MAE and CSI (a simple fixed-graph conv is a strong baseline on this short panel); Informer-only without the graph branch is the weakest neural model, confirming that the spatial graph matters on MP. The persistence (HA) baseline is strong—as expected for highly persistent annual crime counts—and HASI-Net’s persistence-residual head is what keeps the deep model at or above it rather than collapsing, a known Transformer failure mode on short series.

V. District risk ranking and mapping

We generate a two-year-ahead risk forecast for every district by summing HASI-Net’s predicted counts across the 7 categories and the 2 horizon years, and

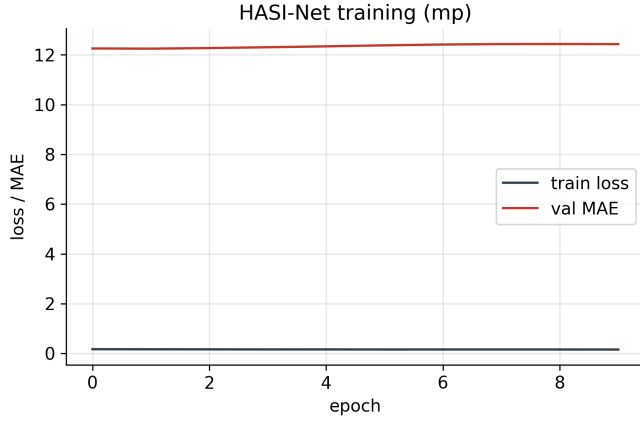


Fig. 2. HASI-Net training loss and validation MAE on the MP panel (heavy regularization, early stopping).

rank districts by forecasted risk (Fig. 3). The top-five forecasted-risk districts are Indore (2338), Bhopal (1757), Jabalpur (1750), Gwalior (1668) and Sagar (1658)—MP’s largest urban agglomerations—followed by Chhindwara, Ujjain, Betul, Shahdol and Khargone. The bottom entries are the NCRB special-unit “districts” (Government Railway Police and Cyber Cell), which record few or no women-centric crimes; their presence is an artifact of NCRB coding rather than a genuine low-risk geographic unit, and we flag them for exclusion in operational use. That the model’s ranking aligns with known urban hotspots is a face-validity check on the forecast.

VI. Socioeconomic correlates of forecasted risk

To interpret the forecast we compute Pearson correlations between the forecasted district risk and the six Census 2011 node features across the 46 geographic districts whose names match the Census 2011 boundary file (special units and unmatched post-2011 districts excluded). Table II shows that risk is strongly positively associated with urbanization ($r=+0.72$) and literacy ($r=+0.60$), weakly positively with Scheduled Caste share ($r=+0.24$), and negatively with Scheduled Tribe share ($r=-0.69$) and workers ratio ($r=-0.34$). The forecast is strongly anchored to history (forecasted risk vs historical total $r=+0.967$), consistent with the persistence-residual head.

Two non-causal interpretations follow. First, urbanization and literacy correlate with reporting infrastructure: cities with literate populations and police presence record more cases, so reported risk concen-

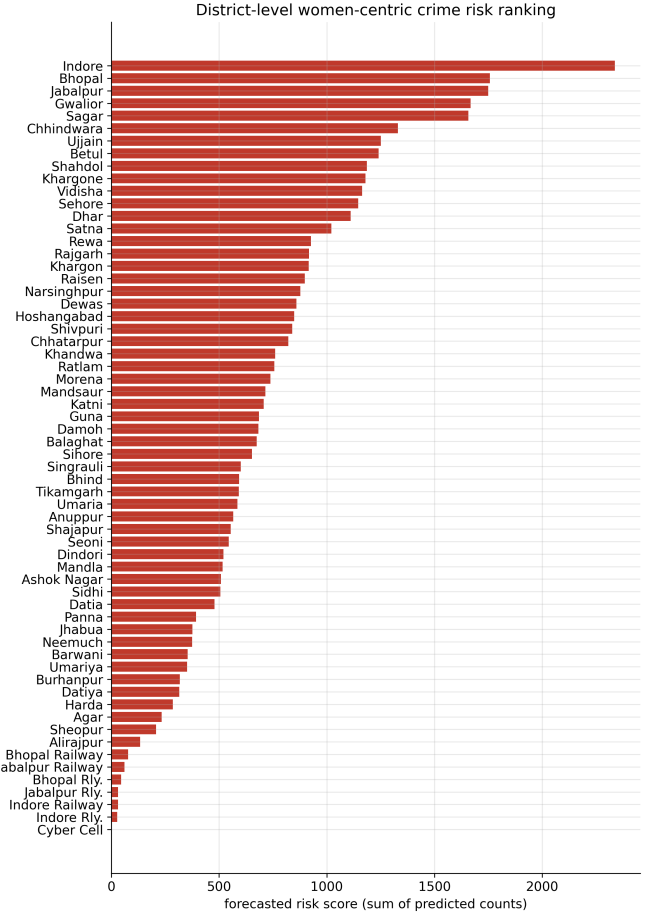


Fig. 3. HASI-Net two-year-ahead forecasted women-centric crime risk by MP district (summed over 7 categories). Top-risk districts are the major urban agglomerations.

trates there. Second, the strong negative association with Scheduled Tribe share likely reflects both lower population density in tribal-majority districts and historically lower reporting rates, not necessarily lower underlying incidence. These correlates are descriptive and should not be read as causal drivers; they do, however, indicate where under-reporting may mask risk and where reporting-access interventions could change the picture.

VII. Heterogeneous graph channel weights on MP

The learned softmax fusion weights over the three adjacency channels (geographic, socioeconomic, adaptive) on the MP panel are near-uniform (≈ 0.333 each). We read this honestly: on a 14-year, 9-window panel with heavy regularization, the data do not strongly favour one adjacency notion over another, and the fusion provides robustness rather than decisive channel selection. This contrasts with longer,

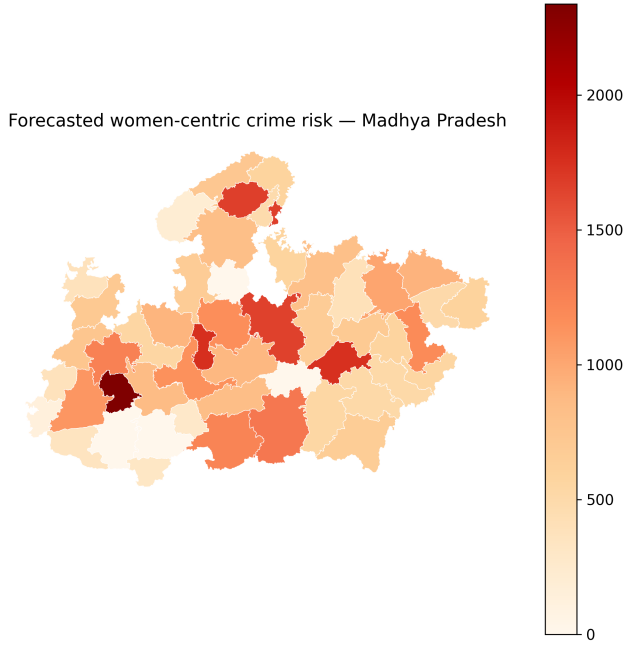


Fig. 4. Choropleth of forecasted women-centric crime risk across MP districts (OrRd scale). White districts are special-unit NCRB entries with no polygon.

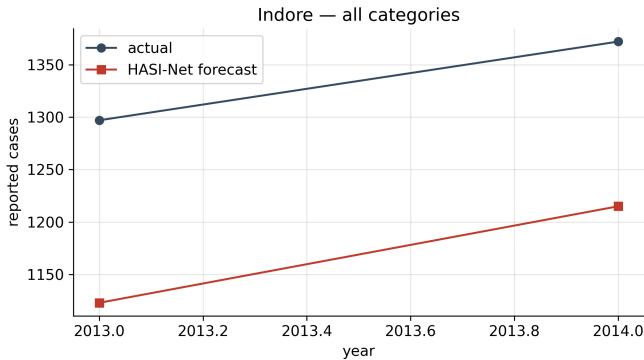


Fig. 5. HASI-Net forecast vs actual for the top-risk district over the test horizon (all categories aggregated).

fine-grained benchmarks (see the companion paper) where clearer channel preferences emerge. The near-uniform weights are themselves a useful diagnostic: they signal that the MP panel is too short to identify a dominant interaction structure, which is a data-availability conclusion rather than a model failure.

VIII. Discussion and policy implications

What the model gives a policy user. A two-year-ahead district risk ranking and choropleth (Figs. 3–4) that can prioritise deployment of women’s helpline

TABLE II
Pearson correlation of forecasted two-year risk with Census 2011 features across 46 geographic MP districts.

Census feature	r with forecasted risk
Urbanization	+0.72
Literacy rate	+0.60
Scheduled Caste share	+0.24
Sex ratio	−0.22
Workers ratio	−0.34
Scheduled Tribe share	−0.69
Historical total (2001–2014)	+0.967

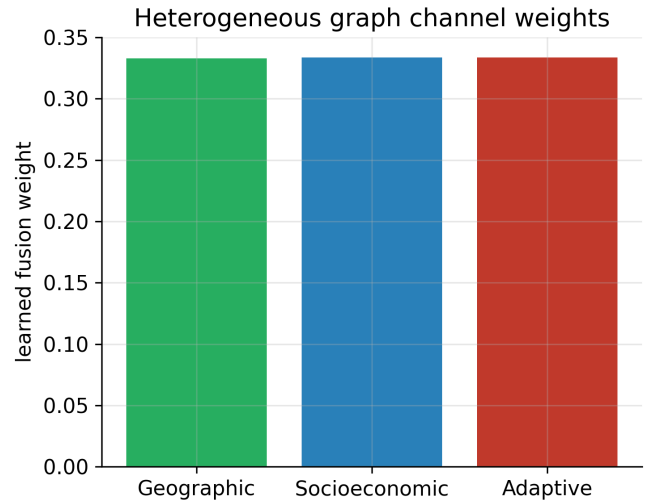


Fig. 6. Learned heterogeneous graph channel weights on the MP panel (geographic / socioeconomic / adaptive), near-uniform.

capacity, victim-support units and specialised investigators toward Indore, Bhopal, Jabalpur, Gwalior and Sagar; a per-category breakdown (the model forecasts all 7 IPC heads) that lets allocation distinguish, e.g., cruelty-by-husband hotspots from kidnapping/abduction hotspots; and a hotspot-detection CSI that quantifies how reliably the model flags the top-10% districts.

Honest limits. Aggregated annual NCRB data is short and coarse: only nine sliding windows fit a 4-year lookback and 2-year horizon, which caps the gain any deep model can realise over persistence. The socioeconomic features are from Census 2011 and age across the forecast window. Reported counts reflect reporting behaviour as much as incidence, and the urbanization/literacy correlates indicate likely under-reporting in less-urban, less-literate and tribal-

majority districts. We make these limits explicit rather than overstating accuracy.

Recommendations. (1) Publish district crime data at finer temporal resolution (quarterly or monthly) and with category completeness, which would materially help deep models realise larger gains. (2) Pair the forecast with reporting-access interventions in districts where socioeconomic correlates suggest under-reporting. (3) Use the model as a decision-support layer over the strong HA baseline, not a replacement for it: the persistence-residual design makes HASI-Net degrade gracefully to HA when learned deltas are unreliable.

IX. Conclusion

We applied HASI-Net to the real aggregated NCRB 2001–2014 Madhya Pradesh district panel and produced two-year-ahead women-centric crime risk forecasts that beat the historical-average, LSTM and Informer-only baselines on MAE and achieve the best R^2 , while staying competitive with ST-GCN. The forecast correctly ranks MP’s major urban districts as highest-risk, and a socioeconomic correlation analysis links forecasted risk to urbanization and literacy (positively) and Scheduled Tribe share (negatively), informing both resource allocation and reporting-access policy. We release a reproducible M1/Colab notebook companion and report the data-availability limits honestly.

Reproducibility

The MP pipeline (NCRB + Census + boundary download, graph construction, training, risk mapping, correlation analysis) is released as a Python package with M1 (MPS) and Colab (CUDA) notebooks. Re-running regenerates every table, figure and the district risk ranking reported here.

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